WIL Project - Milestone 1

Course Code: COSC2669

Group ID: WIL PROJECT - 48

Project Name: Team Nexus

Trello board: https://trello.com/b/xkxMU9td/team-nexus-wil-project-48

GitHub repository: https://github.com/SajilRajendran/Team-Nexus-WIL-PROJECT-48

Project Aim Statement

We are building a Retrieval-Augmented Generation (RAG) chatbot that helps residents and prospective movers understand planning and public transport changes under Victoria's Train and Tram Zone Activity Centres Program, making complex government planning documents accessible through simple, everyday questions.

User Stories

- As a renter considering a new suburb, I want to ask what housing and planning changes are happening near a specific train or tram station, so that I can decide if it's a good place to live.
- As a current resident, I want to find out what developments are planned near my street, so that I understand how my neighbourhood might change.
- As an international student new to Melbourne, I want to compare how well different Activity Centres are served by public transport, so that I can choose a well-connected place to live.

Group Members

Student ID	Full Name	Proposed Role / Skills	% Contribution
S4175707	Harshavardhen Gopal	Evaluation & Testing Lead	16.67
S4193238	Karthic Raj Ayyappan	LLM / Generation Engineer	16.67
S4216604	Rohit Devaraj	Data Preprocessing & Retrieval Engineer	16.67
S4175663	Sajil Rajendran	Domain Lead, Frontend/Demo Developer & Project Coordinator.	16.67
S4139968	Sheryas Senthilkumar	Data Preprocessing & Retrieval Engineer & Report Writer	16.67
S4183343	Trivikram Mohanraj	Data Collection & Report Writer	16.67

Progress So Far

Our team selected our topic - planning and transport information access for Victoria's Train and Tram Zone Activity Centres after evaluating several domains like education, health, urban planning, and public transport against the availability of manageable, public, and well-justified data. We identified a small set of official Planning Victoria documents to use as our knowledge base and confirmed no comparable public-facing solution currently exists for this program.

As part of our early implementation work, our team has been testing Ollama to understand how to set up and run a free, local large language model, including exploring available models and their response behaviour. In parallel, we are studying Walert's own RAG architecture and evaluation approach in detail to guide our implementation decisions. Our document retrieval work is currently in progress: we are exploring how to structure, chunk, and search our chosen Planning Victoria documents effectively. This groundwork is directly informing our next steps of building a working retrieval pipeline and connecting it to Ollama for answer generation.

Plan for the Next Three Weeks

Week	Focus
Week 1	Finish collecting and cleaning source documents for our chosen Activity Centre, complete the chunking and vector database retrieval pipeline; finish connecting Ollama for answer generation. Submit Milestone 1.
Week 2	Connect retrieval and generation into one working end-to-end prototype, build the evaluation test set (known/inferred / unanswerable questions) and run our first evaluation pass.
Week 3	Refine the pipeline based on evaluation results, build and polish the demo interface, prepare a progress update for our mentor and begin scoping the next milestone.

Bottlenecks and Risks

- Official planning documents are dense and technical, which may complicate chunking and retrieval quality.
- Running a free local LLM may be slow on some team members' hardware; the team may need a shared setup or a smaller model.
- Designing a fair set of unanswerable test questions requires careful, team-wide input rather than one person's judgement.

References

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